

Beyond the Rate: Information Content of FOMC Forward Guidance Language

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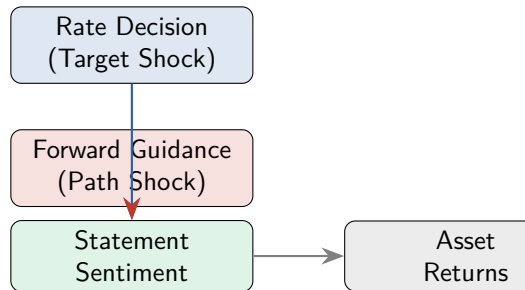
The Puzzle

Central question:

Does FOMC statement language convey information **beyond** the rate decision?

Why it matters:

- Central banks increasingly rely on **forward guidance** as a policy tool
- Markets respond to *what the Fed says*, not just *what it does*
- The “information channel” (Romer & Romer, 2000) suggests FOMC statements reveal the Fed’s superior economic assessment



- **Kuttner (2001)**: 30-min FF futures window → isolate monetary policy surprises
- **GSS (2005a)**: Decompose surprises into **target** + **path** factors
- **Nakamura & Steinsson (2018)**: High-frequency identification of monetary non-neutrality
- **Campbell et al. (2012)**: Delphic vs. Odyssean forward guidance
- **Jarociński & Karadi (2020)**: Separate policy shock from information shock
- **Hansen et al. (2018)**: Computational linguistics for FOMC transparency

Our contribution:

- ① First to directly link **GSS target/path shocks** to **FOMC statement sentiment**
- ② Expanded CB dictionary (97 hawkish + 106 dovish terms, fully disjoint)
- ③ Formal test of the **information channel** via text analysis

Monetary Policy Shocks

- Acosta (2022): 220 FOMC meetings, 1995–2022
- **Target shock**: surprise in current FF rate
- **Path shock**: surprise in future rate path
- Extension: USMPD (SF Fed) → 2026

Market Data

- CRSP VW/EW/S&P 500 via WRDS
- 117 meetings with complete overlap

FOMC Statement Sentiment

- 164 statements, 2006–2025
- Expanded CB dictionary:
 - 97 hawkish terms
 - 106 dovish terms
 - 50 bigram phrases
 - Fully disjoint
- Score = $\frac{H-D}{H+D+\epsilon}$
- Validated: $r = 0.29$ with rate change

Four hypotheses:

Hypothesis	Prediction	Specification
H1	Shocks predict sentiment	$S_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H2	Shocks predict returns	$R_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H3	Path > Target for sentiment	$ \beta_2 > \beta_1 $ in H1
H4	Stronger during FG period	Interaction with FG dummy

Estimation: OLS with Newey-West (lag=1) standard errors

Sample: 117 FOMC meetings, January 2006 – July 2022

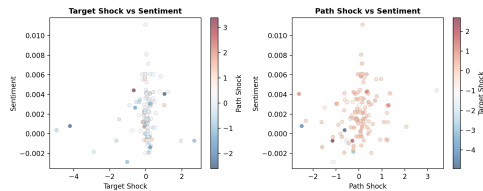
H1: Sentiment and Monetary Policy Shocks

	Coefficient	p -value
Target shock	0.000290	0.062*
Path shock	0.000469	0.047**
R^2	4.06%	
N	117	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Key finding: Path shock is significant at 5%, target at 10%.

But: Wald test $\chi^2 = 0.19$, $p = 0.66$ — cannot reject $\beta_1 = \beta_2$.



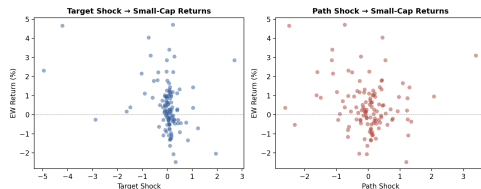
H2: Asset Returns and Shocks

Asset	β_T	p_T	β_P	p_P
CRSP VW	-0.435	0.111	-0.186	0.398
CRSP EW	-0.449	0.044**	-0.174	0.421
S&P 500	-0.391	0.158	-0.179	0.395

Returns in percentage points; $N = 117$

Findings:

- Target shock: significant for EW (small-cap), not VW
- Path shock: **not significant** for any asset
- Small-cap sensitivity consistent with literature



H3: The Information Channel

Prediction: Path shock has larger effect on sentiment than target shock.

- $|t_{\text{path}}| = 2.012 > |t_{\text{target}}| = 1.887$
- Path significant at 5%, target at 10%
- **But:** Wald test $H_0: \beta_{\text{target}} = \beta_{\text{path}}$
 - $\chi^2 = 0.19, p = 0.66$
 - Cannot reject equality

Interpretation: *Suggestive* evidence consistent with the information channel, but **not definitive proof** that the path effect dominates.

Why the modest R^2 (4.06%):

- 96% of sentiment variation is unexplained by shocks
- Likely reflects: economic data releases, institutional inertia, other Fed communication channels

Sample	R^2	$p(\text{Target})$	$p(\text{Path})$	N	Period
Full (Acosta)	4.06%	0.062*	0.047**	117	2006–22
No COVID	4.10%	0.075*	0.042**	115	2006–22
Post-2010	2.02%	0.369	0.108	97	2010–22
Extended	1.65%	0.419	0.058*	163	2006–26

Key takeaways:

- COVID exclusion: virtually unchanged → not driven by outliers
- Post-2010: neither shock significant → ZLB period weakens the signal
- Extended sample (USMPD): path still significant at 10% through 2026

Regime-Specific Results

Period	N	R^2	$p(\text{Target})$	$p(\text{Path})$
Pre-ZLB (2006–08)	8	20.5%	0.360	0.403
Crisis (2008–10)	12	9.2%	0.930	0.124
ZLB/FG (2010–16)	48	7.5%	0.249	0.107
Normalization (2016–20)	30	13.3%	0.009***	0.873
COVID+ (2020–22)	19	3.6%	0.355	0.953

Caution: small N within each regime. Results are suggestive.

Pattern: Target dominates during Normalization (rate hikes); Path is more relevant during ZLB/FG and Crisis periods.

Limitations & Future Work

Limitations:

- ① **Low R^2 :** Shocks explain only 4% of sentiment variation
- ② **No control variables:** No lagged sentiment, macro controls, or meeting type dummies
- ③ **Dictionary-based sentiment:** Cannot capture context or pragmatics
- ④ **Sample size:** $N = 117$ limits power, especially for sub-sample analysis
- ⑤ **US only:** External validity to other central banks unknown
- ⑥ **Multiple testing:** 4 hypotheses without significance level adjustment

Future directions:

- FinBERT / LLM-based sentiment for better measurement
- Control variables: lagged sentiment, macro indicators
- Cross-country: ECB, BoJ, BoE
- Other channels: minutes, press conferences, speeches

- 1 The **path shock** (forward guidance) is a statistically significant predictor of FOMC statement sentiment ($p = 0.047$)
- 2 The **target shock** is also significant at 10% ($p = 0.062$)
- 3 However, we **cannot formally establish** that the path effect dominates (Wald test $p = 0.66$)
- 4 The evidence is **suggestive** of an information channel, not definitive
- 5 Asset returns respond primarily to the **target shock** (small-cap sensitivity), not the path shock
- 6 Results are robust to COVID exclusion and sample extension (USMPD through 2026)

Takeaway: FOMC language is forward-looking, but monetary policy shocks explain only a small fraction of its variation. The information channel exists but is one of many forces shaping central bank communication.

Thank You

Questions & Discussion

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Appendix: Event Study Methodology

Market Model (Brown & Warner, 1985):

- ① Estimation: $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$
- ② Abnormal Return: $AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt})$
- ③ CAR: $CAR_i = \sum_{t=-T_1}^{T_2} AR_{it}$
- ④ **t-stat:** $t = \frac{CAR_i}{\sigma_{AR} \times \sqrt{N}}$

Important: The denominator is $\sigma_{AR} \times \sqrt{N}$, **not** $\sigma_{AR}/\sqrt{N}$.

- σ_{AR} : std. dev. of abnormal returns from estimation window
- N : number of days in event window
- Common error: using $\sigma_{AR}/\sqrt{N}$ inflates t by factor of N

For cross-sectional tests (average CAR across K assets), use $\sigma_{CAAR}/\sqrt{K}$.

Appendix: USMPD Factor Replication

GSS (2005a) two-step orthogonalized PCA:

- 1 Target = 1st PC of [MP1, FF1–FF4]
- 2 Path = 1st PC of [FF4–FF6, ED2–ED4], orthogonalized against target
- 3 Normalized by regressing on daily 1Y GSW yield change

Factor	Instruments	PC1 Variance	Corr w/ Acosta
Target	MP1, FF1–FF4	87.1%	0.958
Path (orth)	FF4–FF6, ED2–ED4	77.0%	0.970
STMT (single)	MP1, MP2, ED2–ED4	78.5%	0.989 (target+path)

Single-factor STMT replicates perfectly ($r = 1.000$).